

Manepalli Mohan Kumar

Curriculum Vitae — AI Engineer, Medical Imaging

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A results-driven AI Developer with 2 years of experience in **Deep Learning** and **Machine Learning**, including 1 year working on AI for **Medical Imaging**. I focused on **Deep Learning (DL)** for **Medical** and **Computer Vision (CV)** models, **Machine Learning (ML)**, **PyTorch** code building, and debugging errors during training and testing for tasks like **medical image segmentation**, classification, and **image translation**. My interests include **vascular disease screening**, **systemic disease detection**, and **fundus & OCT imaging**.

EDUCATION

Bachelor of Technology	2022 — APRIL 2026
<i>Rajiv Gandhi University of Knowledge Technologies</i>	<i>CGPA: 8.5</i>
Pre-University Course (Intermediate / MPC)	2020 — 2022
<i>Rajiv Gandhi University of Knowledge Technologies</i>	<i>CGPA: 9.56</i>
Secondary School Certificate (10th Class)	2020
<i>BSE AP</i>	

- Secured the **State 64th Rank** in the 2020 Polytechnic Entrance Examination immediately following 10th grade.

PUBLICATIONS & PRESENTATIONS

"Energy-Regularized Self-Supervised Denoising for Structure-Preserving OCT Imaging"	MARCH 2026
<i>Optics Letters Journal</i>	<i>Published</i>
Addressed the over-smoothing problem in S2SNet for the OCT dataset by introducing energy regularization and validated improvements through a comprehensive comparison of segmentation performance.	
"Data Driven Approach in Building Consumer-Friendly Business Models Using Machine Learning Techniques"	DEC 2024
<i>IEEE AICECS Conference</i>	<i>Manipal Institute of Technology</i>
Presented a research paper comparing three machine learning models on real-time dataset metrics.	

PROFESSIONAL EXPERIENCE

Technical Staff / TANUH (AICoE) IISc	JUNE 2026 — PRESENT
<i>Vascular Disease Screening Vertical</i>	<i>Bangalore, KA</i>
- Working in the Vascular disease screening vertical to detect systemic diseases from only 4 fundus images per patient, leveraging RETFound, MedGemma 1.5, and Qwen 3.8 27B with LoRA fine-tuning. - Built an agent-based automated data quality check pipeline to validate incoming clinical imaging data at scale. - Developed and operated a data collection and anonymisation website (developer & IT owner) hosted on Google Cloud Platform (GCP) across India for real-time data collection from hospitals. - Administered the team's compute nodes and Open OnDemand software, managing access and workloads for researchers.	
Engineering Intern / TANUH (AICoE) IISc	NOV 2025 — MAY 2026
<i>Medical Assistant and Early Disease Detection</i>	<i>Bangalore, KA</i>
- Working at TANUH (AI Centre of Excellence under the Ministry of Education) to develop AI-driven diagnostics and screening tools for early breast cancer detection and risk prediction. - Developing "Pseudohealth" generative systems to synthesize mammogram images and finetuned MedGemma 1.5 4B to work with mammogram images, transforming clinical care through scalable AI solutions.	
Developer / Video Masking & Annotation Application	AUG 2025 — NOV 2025
<i>Full-Stack AI Deployment</i>	<i>Bangalore, KA</i>
Designed a medical video annotation tool integrating the Segment Anything Model (SAM) for frame-by-frame analysis, deployed via a scalable serverless architecture on Google Cloud Run.	
Research Intern / MIGLab [Link], CDS Department at IISc	MARCH 2025 — OCT 2025
<i>Medical Imaging & Deep learning</i>	<i>Bangalore, KA</i>
- Executed distributed training via torchrun across DGX Spark clusters and optimized model inference through Quantization for real-time serving. - Implemented Flash Attention and Maximum Representative Subsampling (MRS) pipelines for the Federated Tumor Segmentation (FeTS) dataset to train 3D UNet models, optimizing data quality and training performance for large-scale vision tasks. - Developed OCT image denoising algorithms and currently developing model-based Quantitative Susceptibility Mapping (QSM) reconstruction algorithms, focusing on preserving structural details and high-precision accuracy.	
Teaching Assistant / Aster Academy Cohort-5	2025
<i>AI in Healthcare / Medical Image Analysis</i>	<i>Bangalore, KA</i>
Supported instruction and technical experiments for a course on AI in Healthcare focused on medical image analysis.	

- Gained professional experience in applied machine learning, focusing on model internals, data curation, and version-controlled experimentation for industrial use cases.

ACADEMIC PROJECTS

- Deployment / Real-time Median-nerve SegmentationJUNE 2023 — AUG 2023
- Deployed VisTr on Jetson Orin NX using NVIDIA SDK Manager, while optimizing transformer performance with Flash-Attn 2 and scaling training via PyTorch DDP on dual RTX A6000 GPUs.
- Experimentation / Stable Diffusion 3.5 Medium & ComfyUI2024
- Executed inferencing of Stable Diffusion 3.5 to generate high-quality images from text prompts utilizing the ComfyUI node-based interface on Kaggle infrastructure.
- Developer / AgriBot (Collab with ICAR, Pune)JAN 2024 — AUG 2024
- Developed an end-to-end multimodal application for leaf disease detection and crop recommendation, integrating vision capabilities with Arduino, ESP, and Raspberry Pi for real-time inference.
- Developer / Gemini AI-Powered Telegram BotDEC 2023
- Built a multimodal agentic chatbot integrated with Google’s Gemini API, utilizing precise model configuration for domain-specific user query answering.
- Developer / Cloud-Based Automatic Object Finder2023
- Engineered a deep learning system for real-time multimodal object identification, utilizing cloud-streamed data and YOLO architectures.
- Developer / LI-FI Prototype2023
- Developed a hardware-integrated prototype for high-speed wireless data transmission via light, demonstrating technical versatility in hardware-AI systems.

TECHNICAL SKILLS

Programming & Frameworks	Python (PyTorch, PyTorch Lightning, TensorFlow, Scikit-learn, MONAI), MATLAB, OpenCV, C/C++, HF Transformers, ComfyUI
LLMs & Generative AI	Vision Language Models (VLMs) & Large Language Models (LLMs) — basic knowledge with some hands-on experience; Diffusion Models (DDPM), VAEs, GANs, Fine-tuning
DL Architectures	Transformer-based models (ViTs), Flash Attention, UNet, Multi-task & Multi-modal Learning
Imaging Systems & Infra.	Medical Imaging (MRI, CT, Fundus, OCT), DICOM/NIfTI formats, Model Optimization, Edge/Mobile Architectures
CV & Processing	Image/Video Restoration, Enhancement, Denoising, Segmentation, Object Detection, OCR, Feature Extraction
Distributed Training & Infra.	DeepSpeed, torchrun, PyTorch, PyTorch Lightning DDP, GPU Acceleration, bfloat16, DGX Spark Clusters
Inference Optimization	Model Quantization (GGUF), Knowledge Distillation, vLLM, TensorRT
Algorithms & Concepts	Regression, Decision Trees, Random Forest, Gradient Boosting, Clustering, PCA, Neural Networks, CNNs
Experiment Tracking & DevTools	MLflow, TensorBoard, Python Logger, Git CLI
Communication	English, Telugu (Native), Hindi, Technical Documentation

CERTIFICATIONS

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|---|------|--|------|
| • Machine Learning for Engineering: NPTEL / IIT Madras [Link] | 2025 | • Circuits & Electronics 3: edX / MITx [Link] | 2024 |
| • VLSI Design Flow: RTL to GDS: NPTEL / IIIT Delhi [Link] | 2025 | • Geoprocessing using Python: IIRS / ISRO [Link] | 2023 |
| • Electronic Systems Design (PCB): NPTEL / IIT Madras [Link] | 2025 | • Intro to Machine Learning: Kaggle [Link] | 2023 |
| • Python (Basic) Certified: HackerRank [Link] | 2025 | • AI Appreciate & AI Aware (Combined): Intel [Link] | 2023 |
| • Essential Mathematics for ML: NPTEL / IIT Roorkee [Link] | 2024 | • Pre-University Course (Intermediate / MPC): RGUKT [Link] | 2022 |
| • Deep Learning Specialist (PyTorch, TensorFlow, Keras): edX / IBM [Link] | 2024 | • Secondary School Certificate (10th Class): BSE AP [Link] | 2020 |

ACTIVITIES

- Regularly analyze Q1/Q2 journals and state-of-the-art (SOTA) research to stay at the forefront of AI advancements.
- Participant: IEEE YESIST’12 Competition [Link]2024
- National Service Scheme (NSS)2021 — 2024